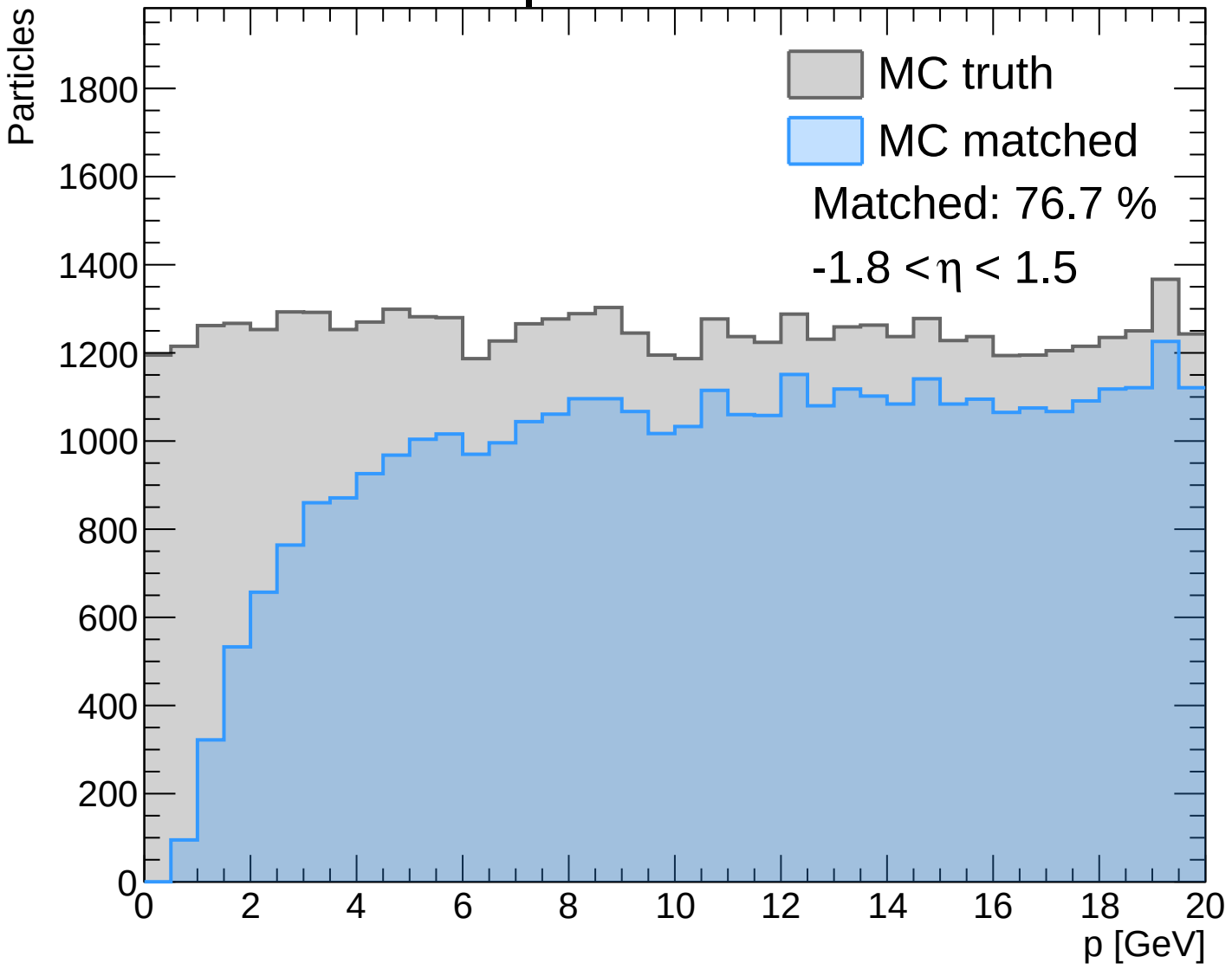
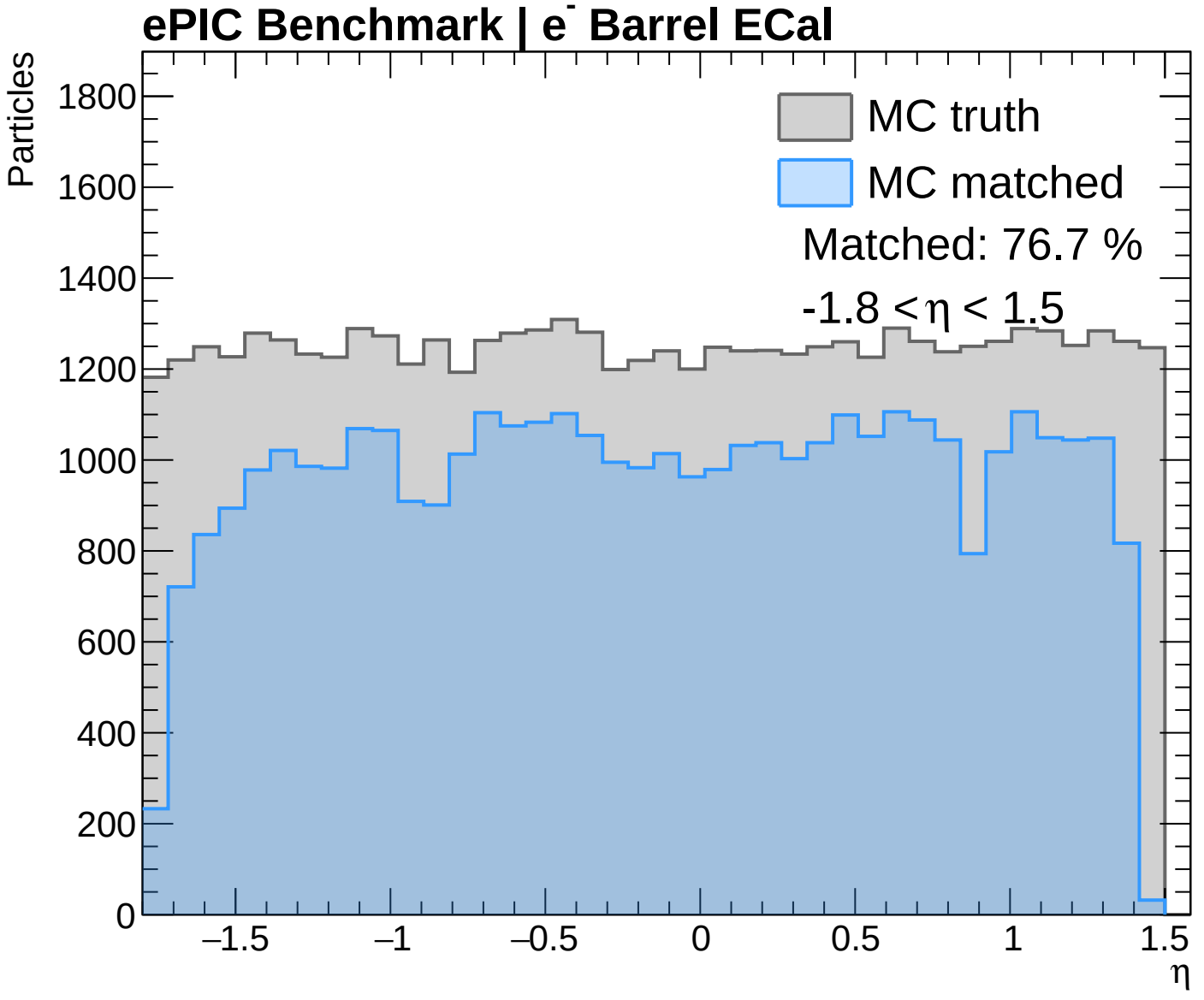
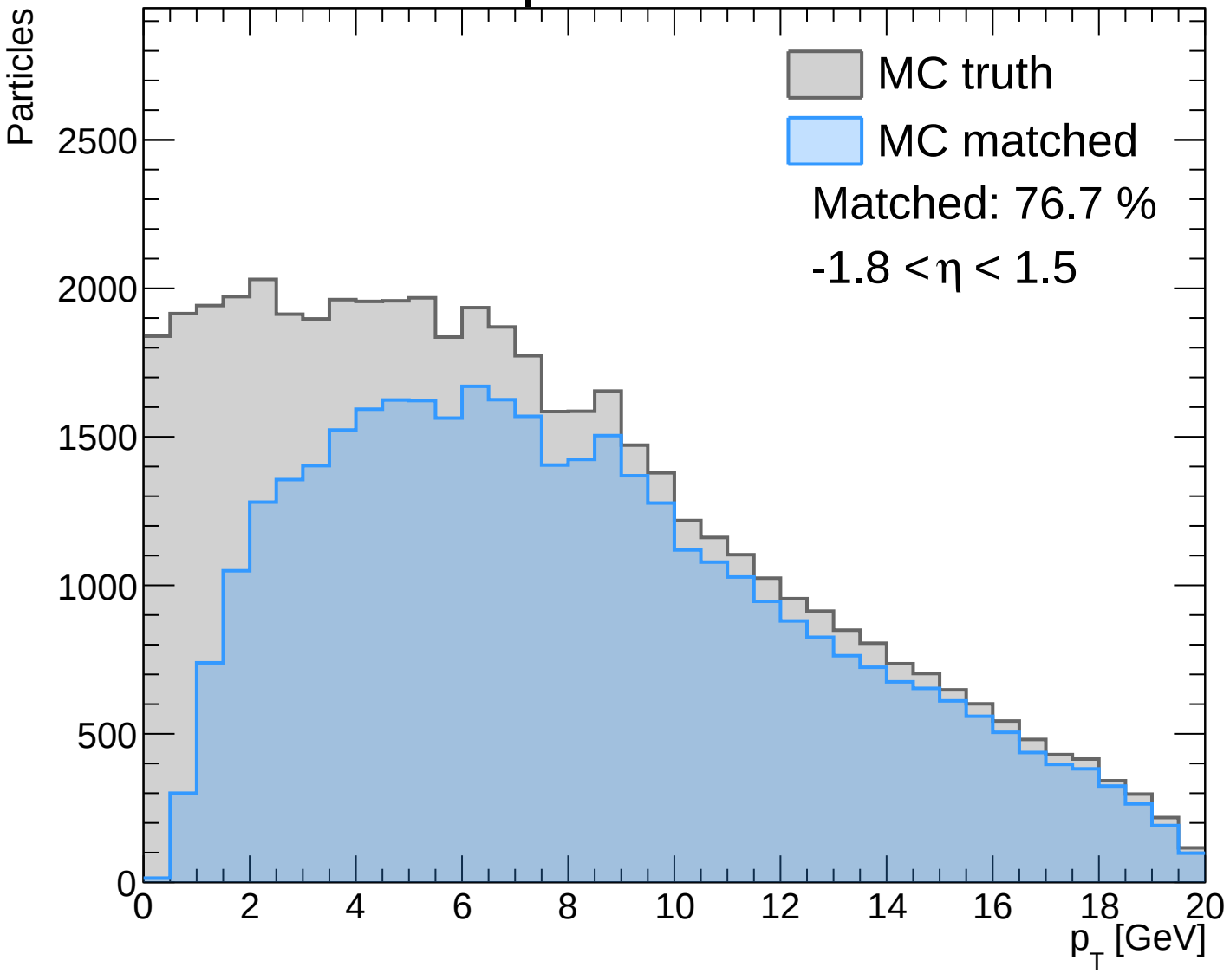
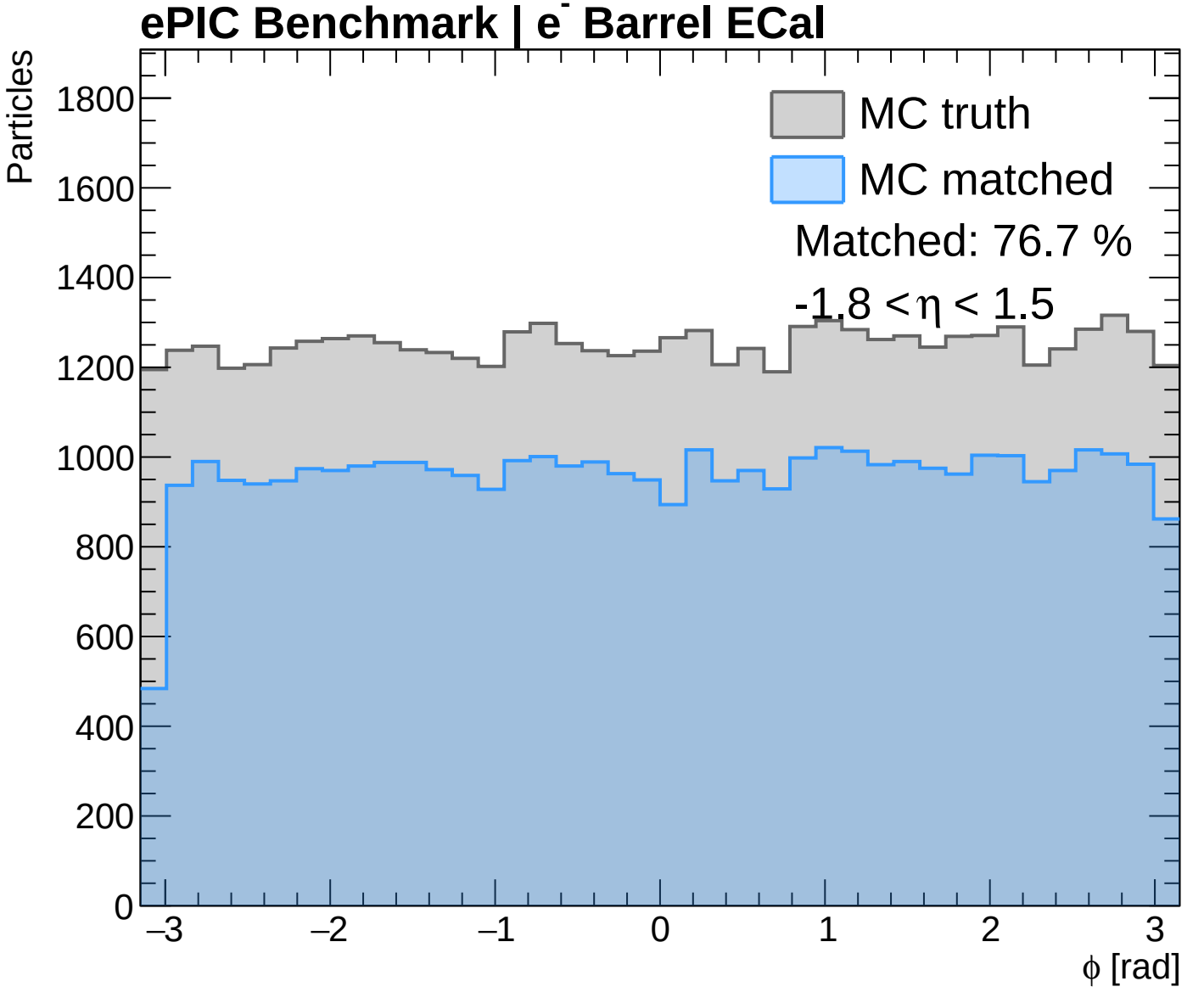
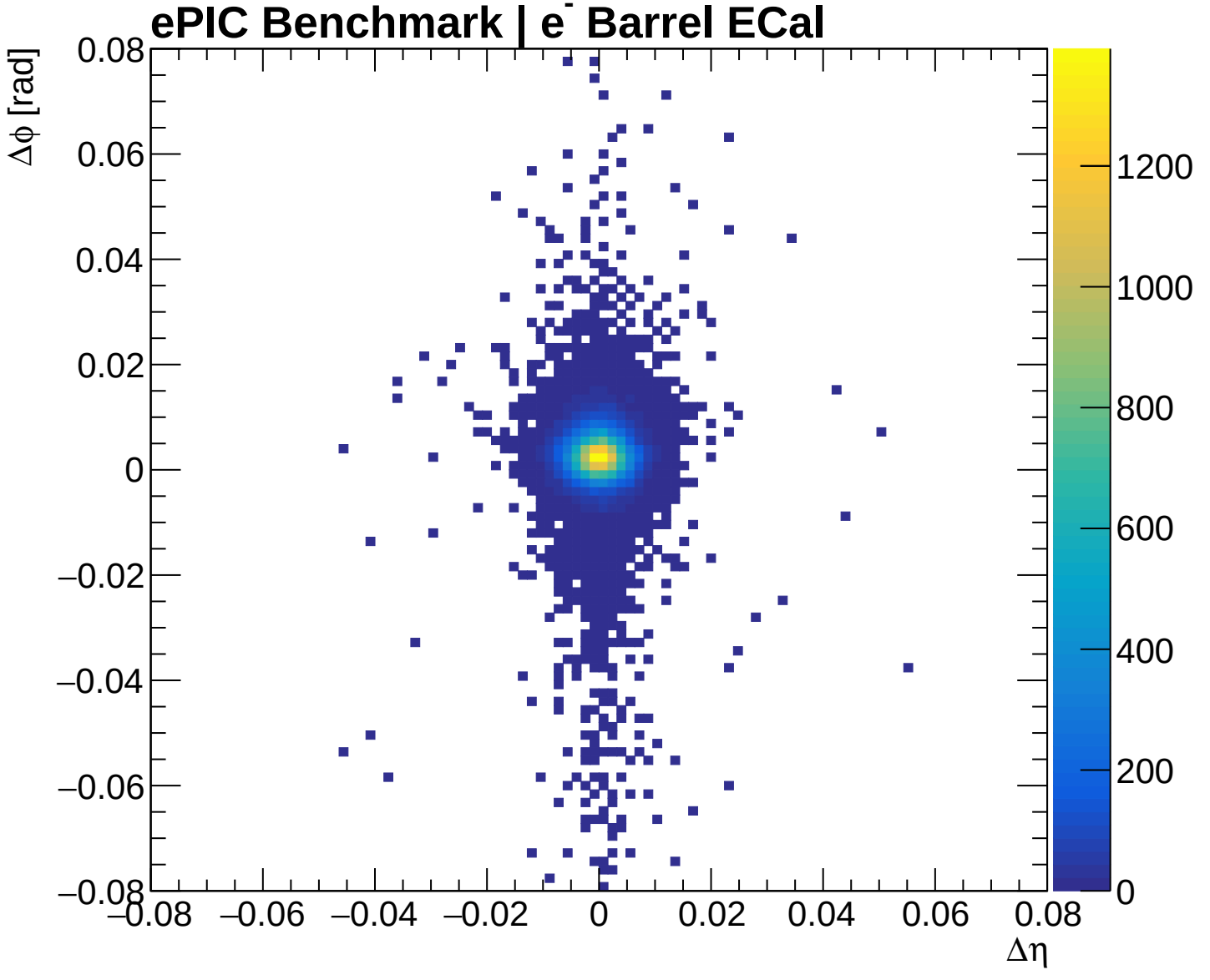
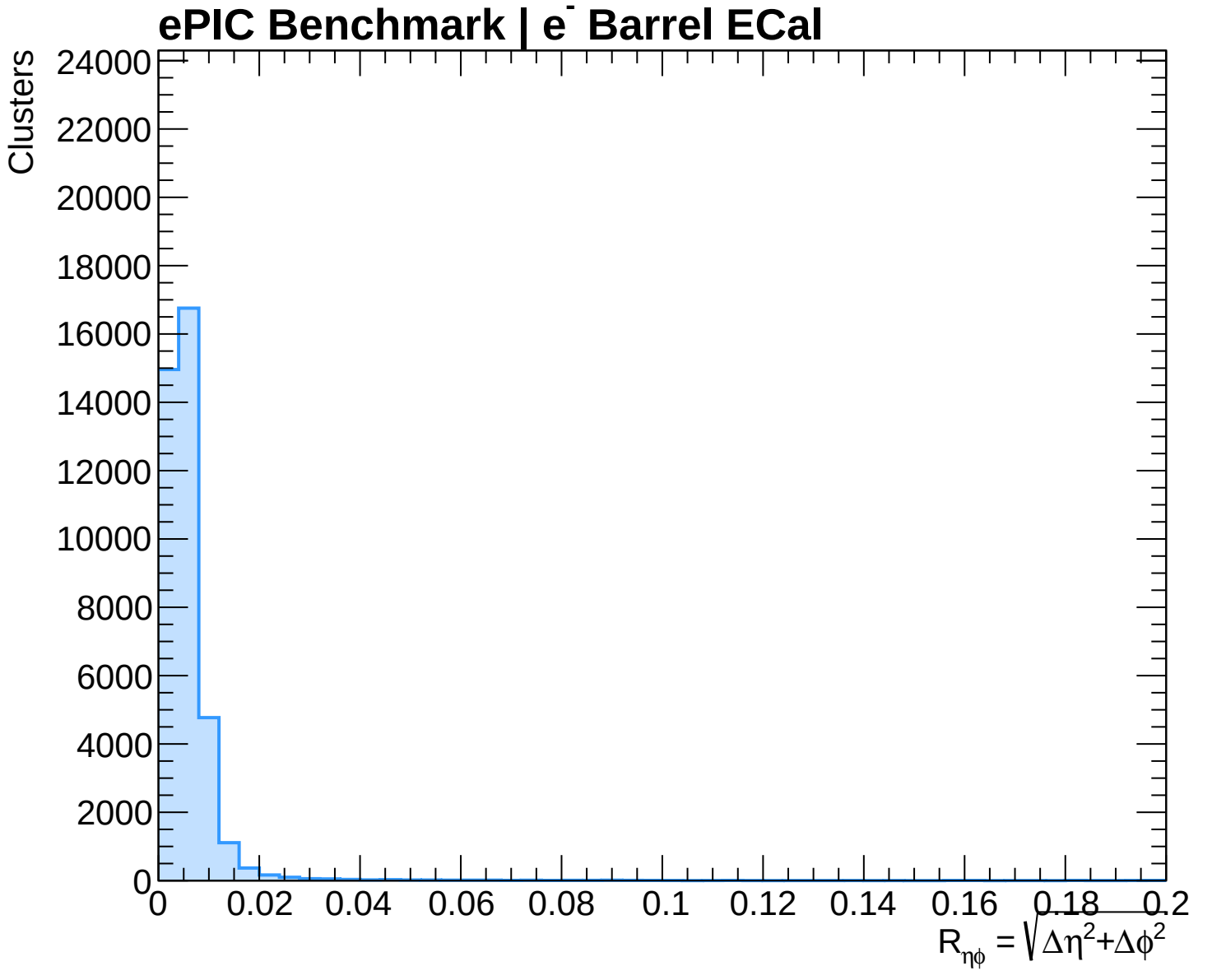

ePIC Benchmark | e- Barrel ECal
Particles
MC truth
MC matched
Matched: 76.7 %
-1.8 < η < 1.5
0
200
400
600
800
1000
1200
1400
1600
1800
0
2
4
6
8
10
12
14
16
18
20
p [GeV]

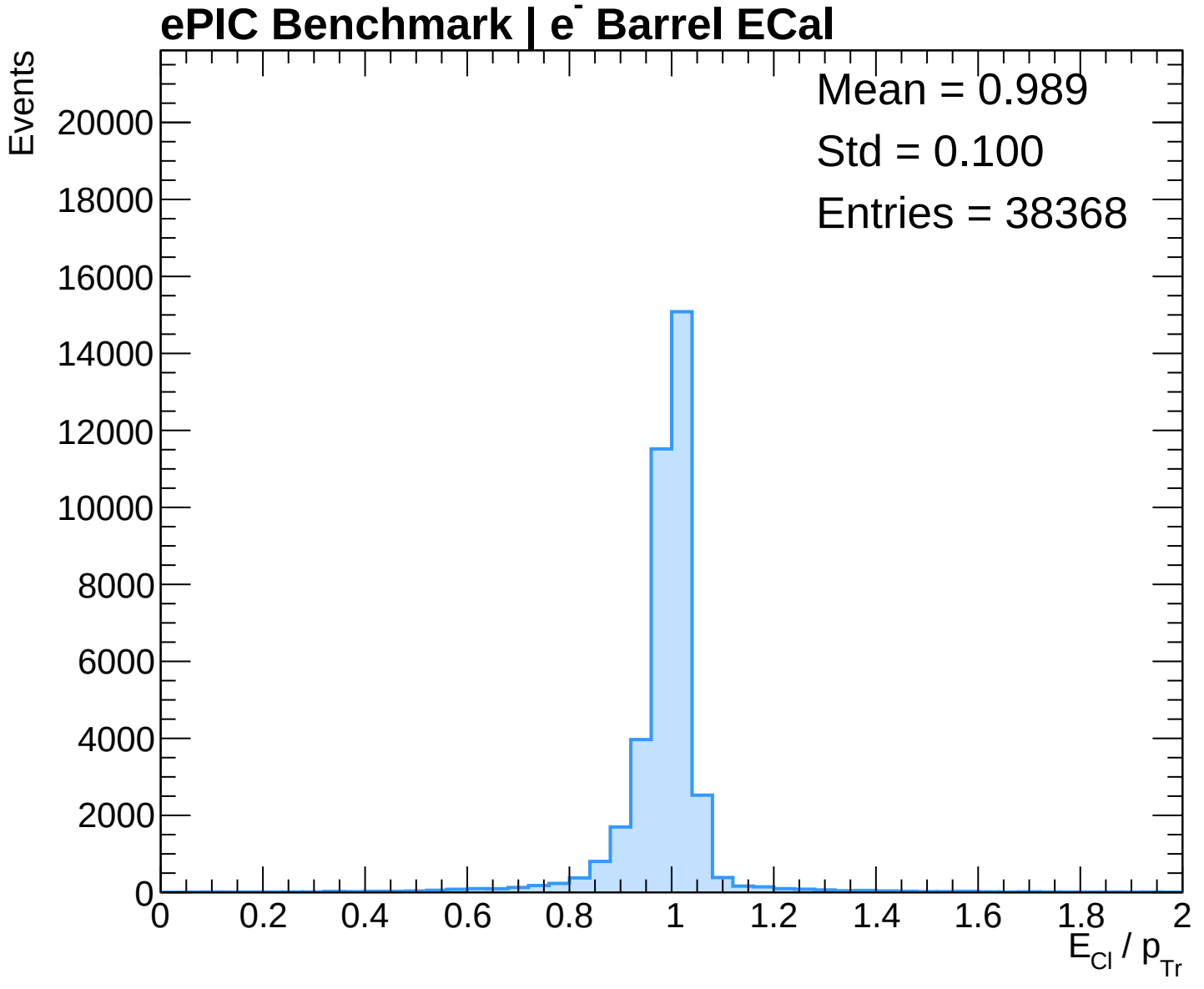

ePIC Benchmark | e- Barrel ECal
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
–1.5
–1
–0.5
0
0.5
1
1.5
η
MC truth
MC matched
Matched: 76.7 %
-1.8 < η < 1.5

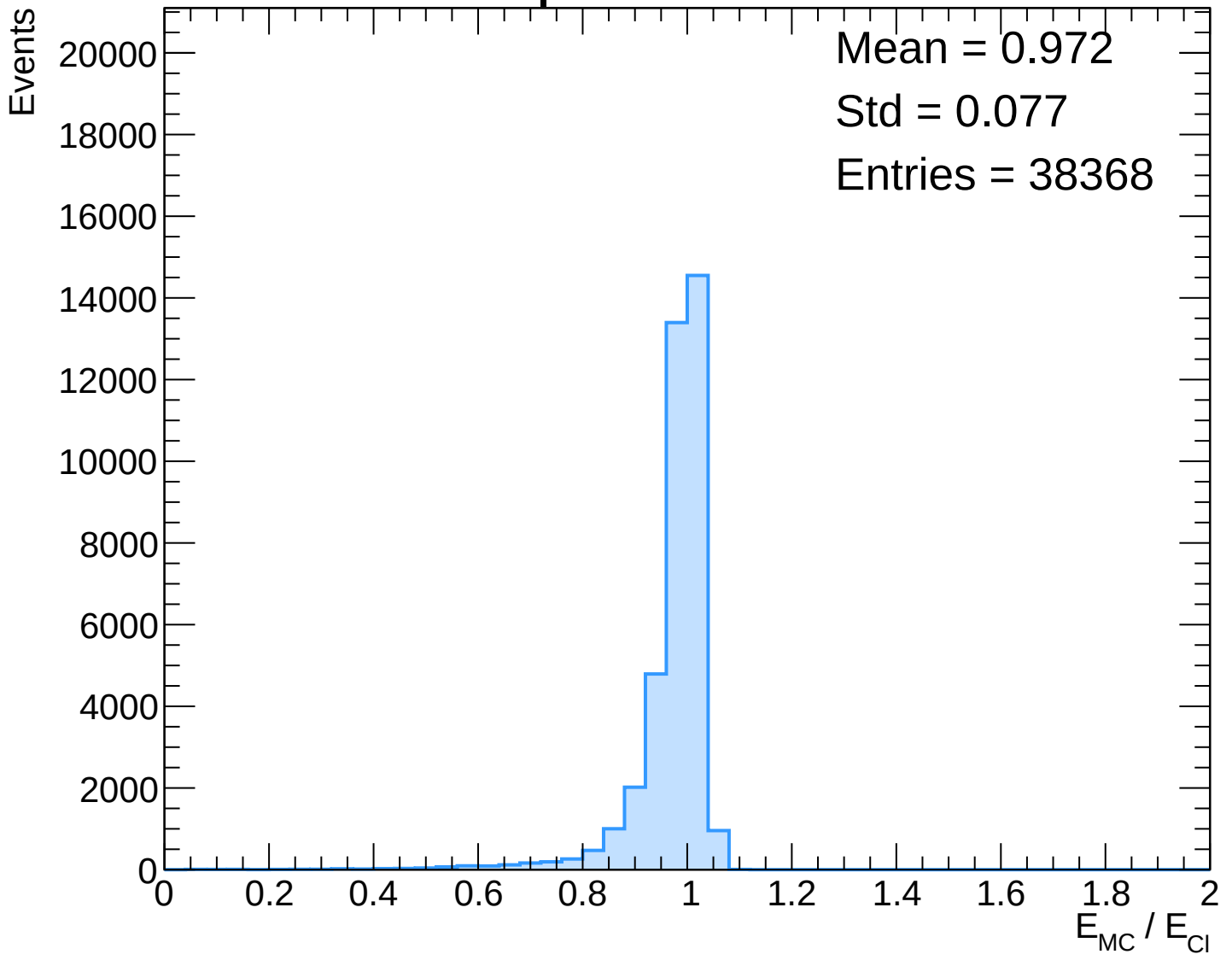

ePIC Benchmark | e- Barrel ECal
Particles
MC truth
MC matched
Matched: 76.7 %
-1.8 < η < 1.5
0
500
1000
1500
2000
2500
0
2
4
6
8
10
12
14
16
18
20
pT [GeV]

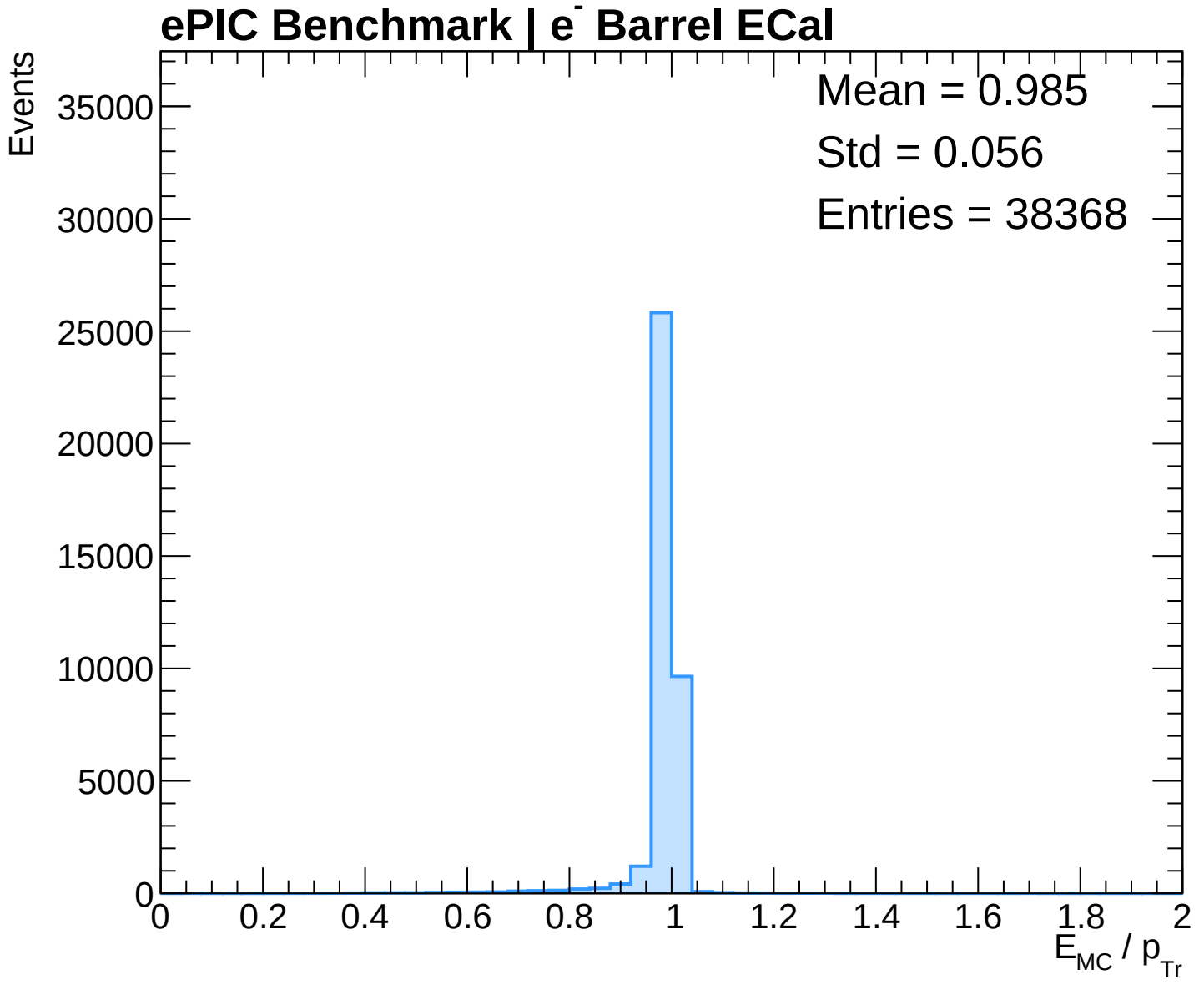

ePIC Benchmark | e⁻ Barrel ECal
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
MC truth
MC matched
Matched: 76.7 %
-1.8 < η < 1.5
–3
–2
–1
0
1
2
3
ϕ [rad]

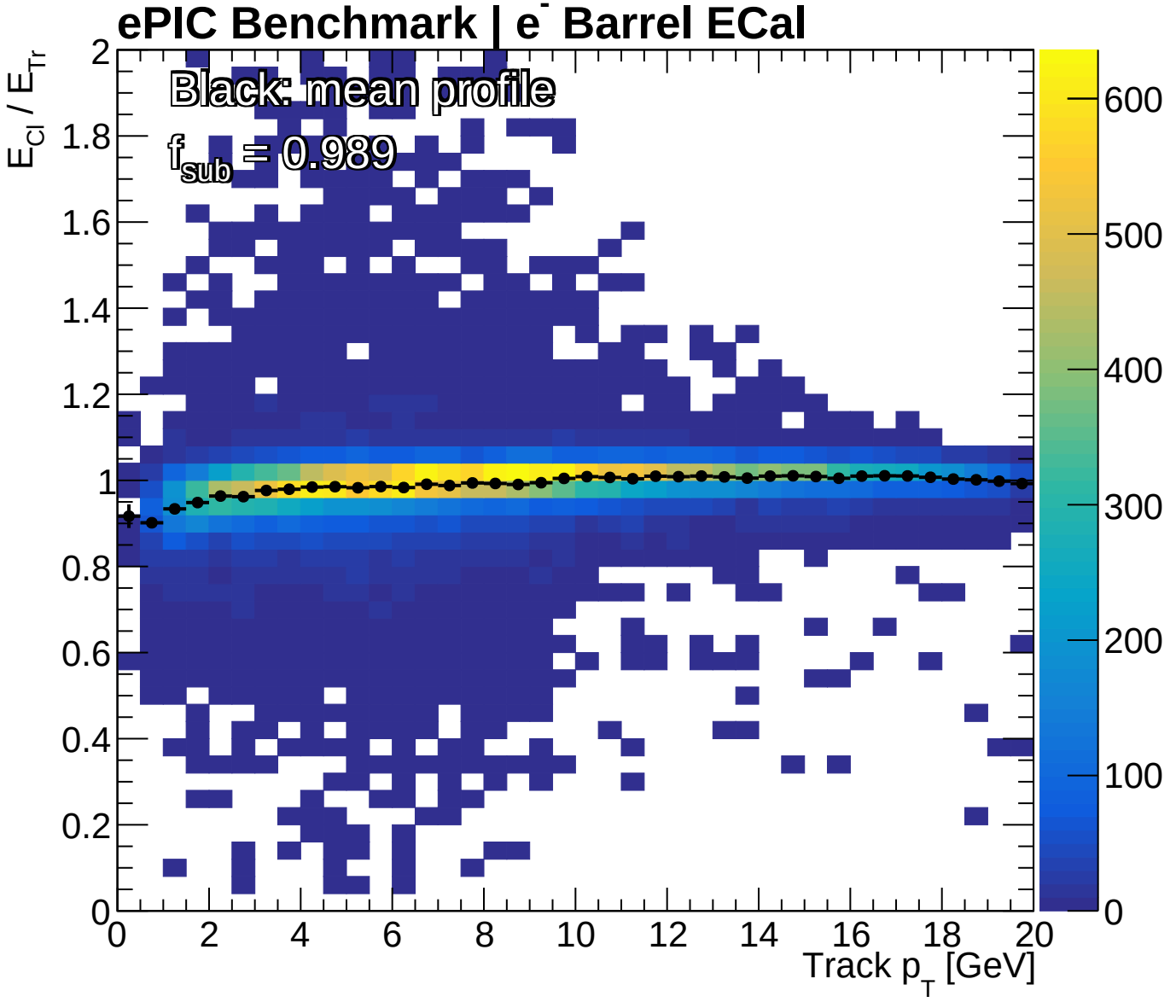

ePIC Benchmark | e⁻ Barrel ECal
Δϕ [rad]
0.08
0.06
0.04
0.02
0
–0.02
–0.04
–0.06
–0.08
–0.08 –0.06 –0.04 –0.02 0 0.02 0.04 0.06 0.08
Δη
1200
1000
800
600
400
200
0

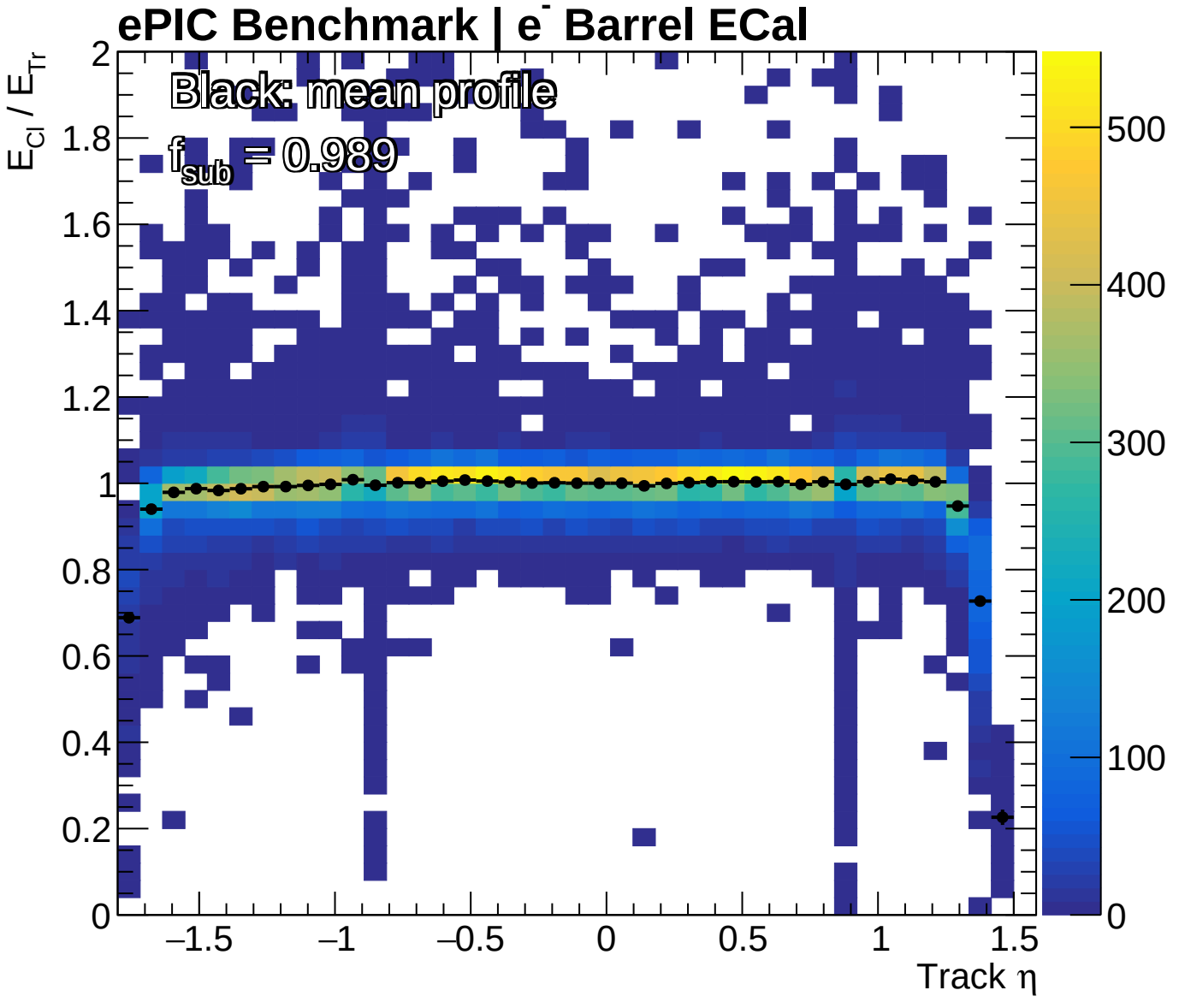

ePIC Benchmark | e- Barrel ECal
Clusters
24000
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
$R_{\eta\phi} = \sqrt{\Delta\eta^2+\Delta\phi^2}$

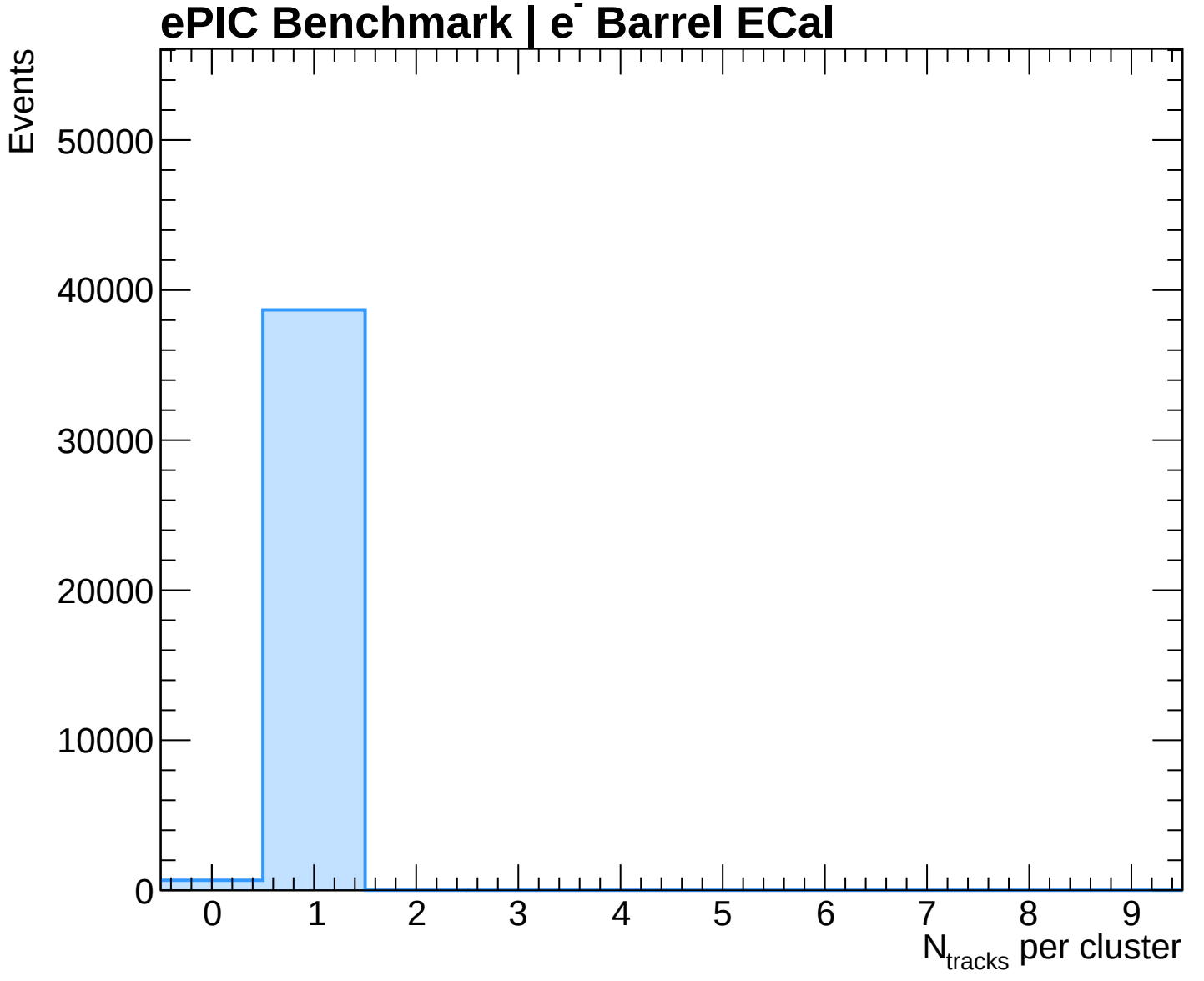

ePIC Benchmark | e- Barrel ECal
Mean = 0.989
Std = 0.100
Entries = 38368
Events
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_Cl / p_Tr

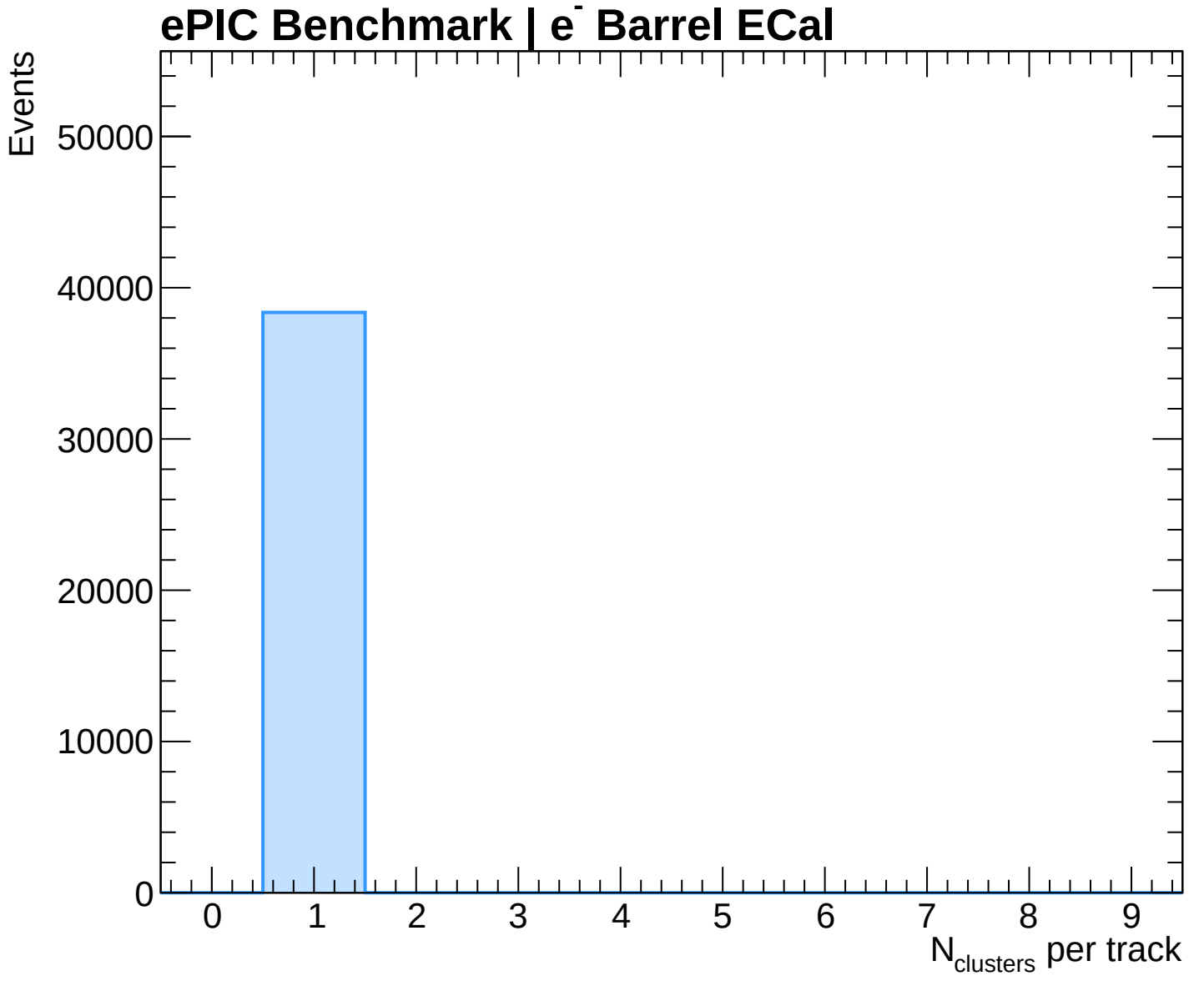

ePIC Benchmark | e- Barrel ECal
Mean = 0.972
Std = 0.077
Entries = 38368
Events
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / E_Cl

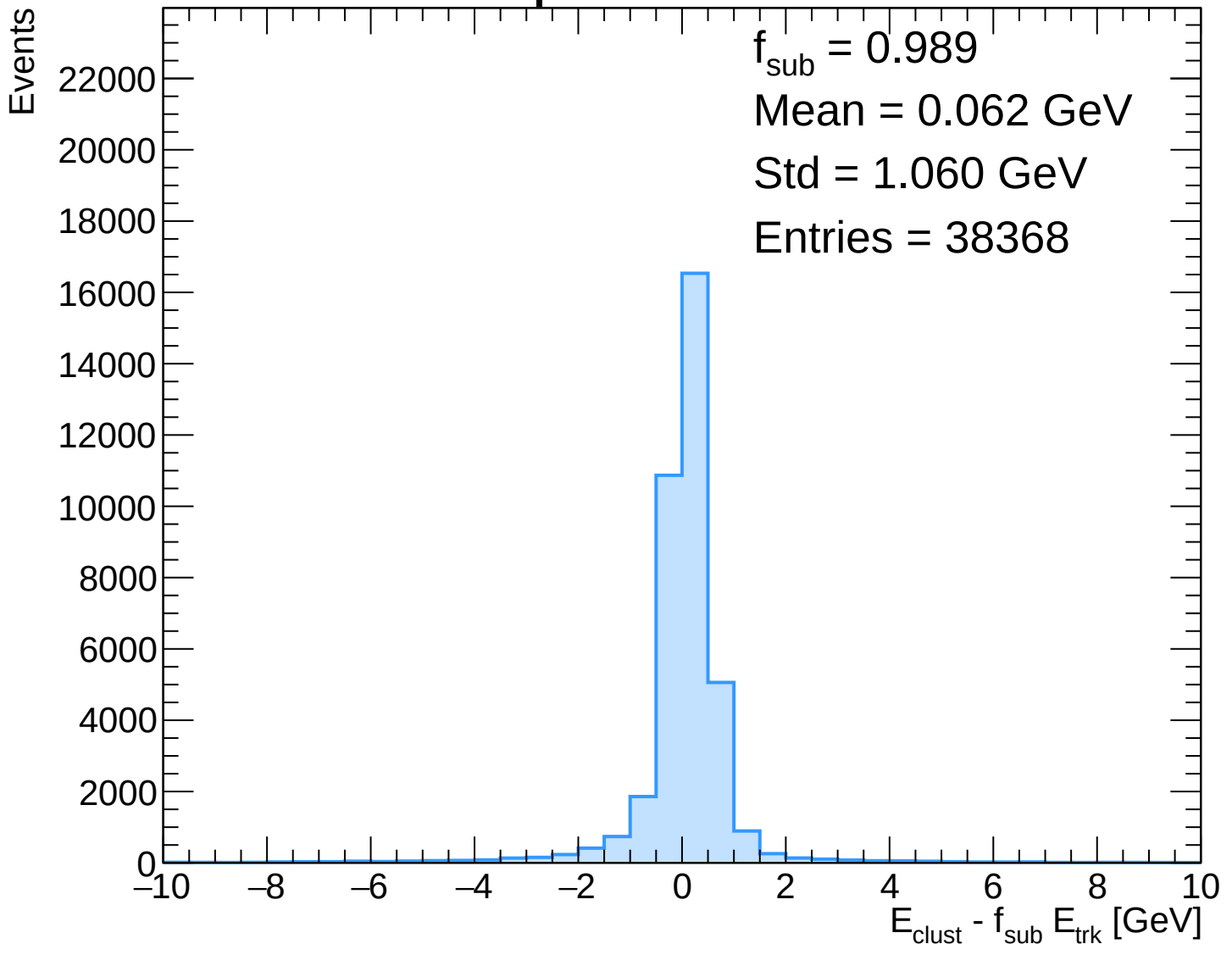

ePIC Benchmark | e- Barrel ECal
Mean = 0.985
Std = 0.056
Entries = 38368
Events
35000
30000
25000
20000
15000
10000
5000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / p_Tr


ePIC Benchmark | e⁻ Barrel ECal
Black: mean profile
f_sub = 0.989
E_Cl / E_Tr
Track p_T [GeV]
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
2
4
6
8
10
12
14
16
18
20
0
100
200
300
400
500
600


ePIC Benchmark | e⁻ Barrel ECal
Black: mean profile
f_sub = 0.989
E_Cl / E_Tr
Track η
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
–1.5
–1
–0.5
0
0.5
1
1.5
500
400
300
200
100
0


ePIC Benchmark | e- Barrel ECal
Events
50000
40000
30000
20000
10000
0
0
1
2
3
4
5
6
7
8
9
N_tracks per cluster


ePIC Benchmark | e- Barrel ECal
Events
50000
40000
30000
20000
10000
0
0
1
2
3
4
5
6
7
8
9
N clusters per track


ePIC Benchmark | e- Barrel ECal
Events
f_sub = 0.989
Mean = 0.062 GeV
Std = 1.060 GeV
Entries = 38368
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
–10
–8
–6
–4
–2
0
2
4
6
8
10
E_clust - f_sub E_trk [GeV]